DAB Failure Problems and CTC CAE Capability

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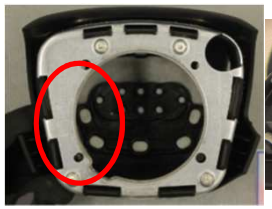
DAB Failure Problems

- DAB failure problems
  - Cover
    - Cover Hinge breakage
    - Deco. Ring slot breakage
    - Hook window breakage
  - Emblem & Deco. ring
    - Nuggets detach
    - Emblem breakage
  - Housing
    - Hook detach
    - Hook Breakage
  - Omega spring
    - Hook (Armature of housing) detach



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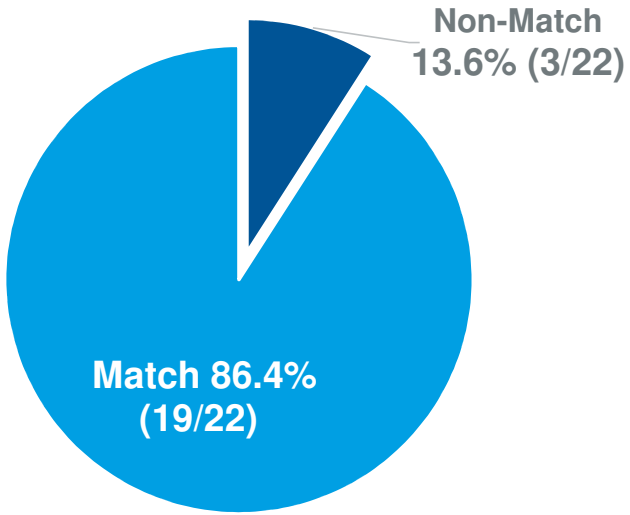
DAB Failure Problems Description

- 2018-2020.09 DAB test results(29 projects) statistics.

Project	AFIS	CAE result	Testing result
Baojun CN202	46708	OK	OK
Chery T1D	66583	OK	OK
Ford CD542	16C07	OK	OK
Geely DC1E	86468	Deco. ring breakage	Deco. ring breakage
Geely DNL-5	86337	OK	OK
Geely FE57B	86425	OK	OK
GLY SX11	86338	OK	OK
Geely SX12	86443	OK	OK
GM 9BYC-S	46558	OK	OK
GM BUI G2XX	46636	OK	OK
GWM A01	86522	OK	DAB Module detached
GWM B01	86485	OK	OK
GWM CH151X	86264	OK	OK
GWM CHB071	66242	OK	OK
GWM CHB071(Illuminate)	66242	Housing window breakage; Emblem nuggets breakage	Housing window breakage; Emblem nuggets breakage
GWM CHB073	86355	OK	OK
GWM CHB131B	86311	OK	OK
GWM P3011	86412	Cover hinge ovetear	Cover hinge ovetear
Haima SG00	86421	OK	Cover wall breakage @ 12H
Baojun CN300M	46667	OK	Housing hook breakage@6H
GWM V61	86488	OK	OK
GM A2XL (Carry over)	46713	OK	Cover wall fusing by inflator@ 12H
GM E2UL (Carry over)	46079	Cover hinge ovetear	Cover hinge overtear
Mazda J59R (Carry over)	16B72	Cover hinge breakage	Cover hinge breakage
PSA E43C (Carry over)	P6650	Emblem nuggets breakage	Emblem nuggets breakage
VW A3 (Carry over)	36192	OK	Emblem slot crack
VW B8L (Carry over)	26559	Cover hinge breakage	Cover hinge breakage
VW V5 (Carry over)	56681	Cover hinge breakage	Cover hinge breakage

CTC CAE Prediction Accuracy



Note: There are 25 projects' tests still not be included.

Carried over from global design, CAEs are done by global, tests are failed in CTC.

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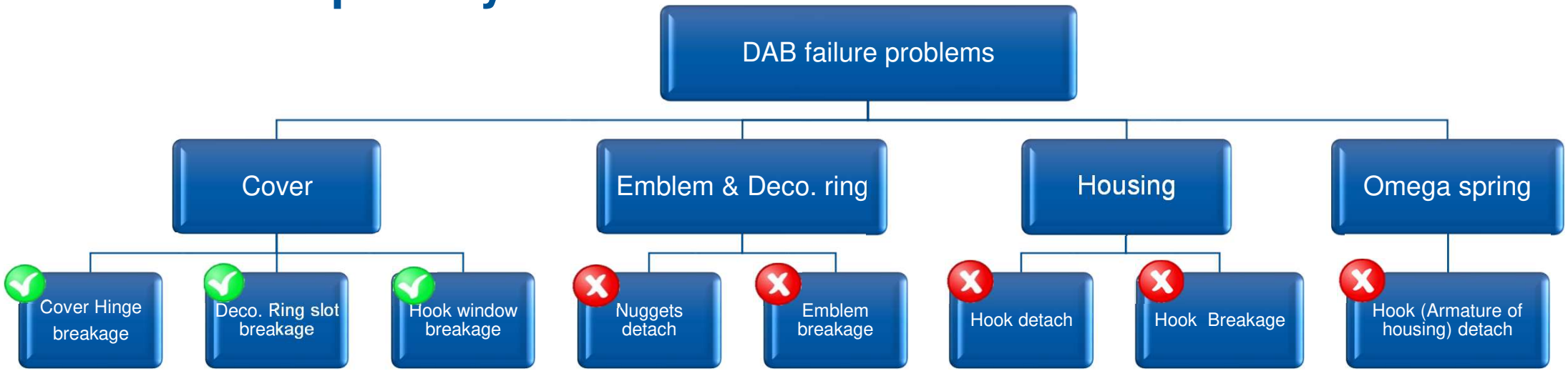
CTC CAE Capability



So far, three cover material card (TT1081B, TA4003BE and TT990) are be calibrated and used in some projects, these cover failure prediction have a better performance.

1. Emblem material(T45-M) is not calibrated;
2. Nuggets weld quality can not be considered;
3. Testing results variation is hard to predict by CAE

1. For steel housing. The residual stress results from assembly process is not considered;
2. For plastic housing. Housing material(PG8) is not calibrated

1. Real friction coefficient of Omega and hook is not considered;
2. Material propertied(POM) of other parts are uncertain.

Temperature influence DAB static deployment tests

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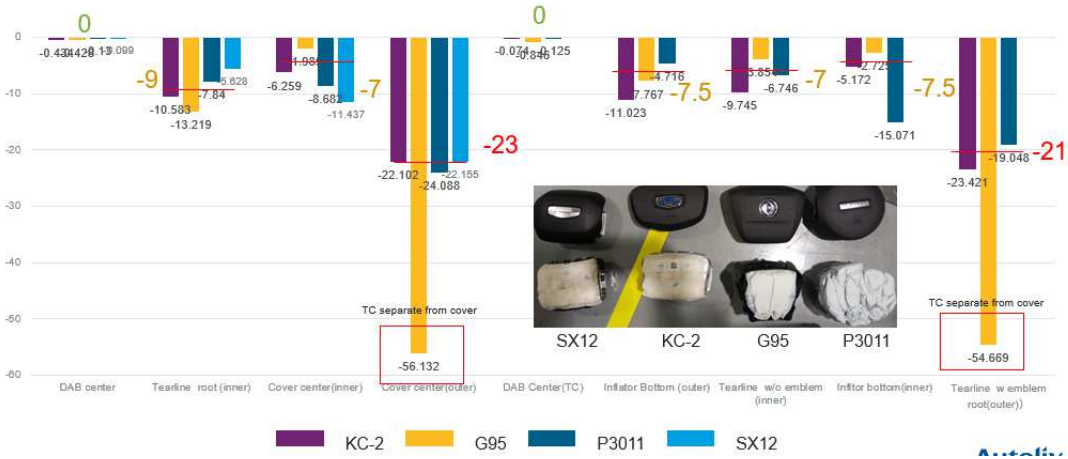
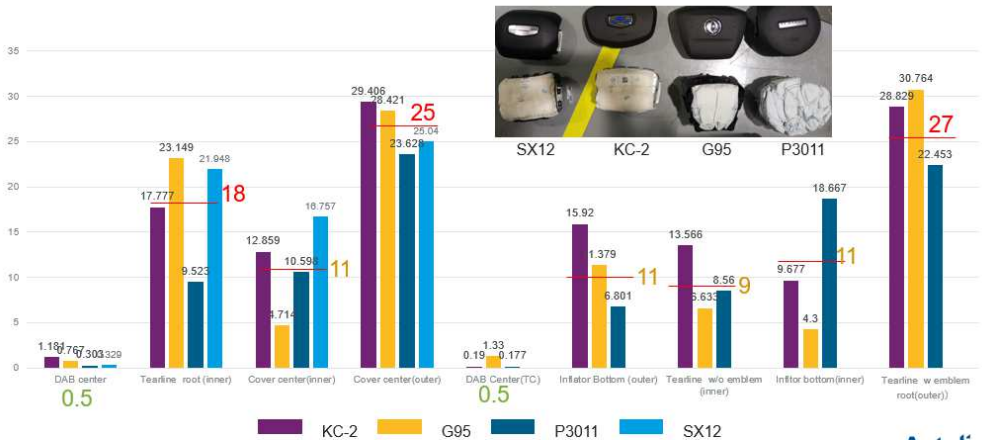
Temperature distribution in some DAB static deployment tests

Because in some high temperature and low temperature static deployment tests, the DAB module must be taken out of environment chamber and deployment after 2 min or 0.5 min in room temperature, the temperature distribution of entire cover module will be change.

The material property differences at different temperature is hard to predicted and considered in CAE.



Temperature distribution after 2 min low temperature (-35 ℃)



Temperature distribution after 2 min high temperature(85 ℃)

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